

## Technical Documentation — FEMA Disaster Dashboard

### Data Sources

Primary datasets used:

- us\_disaster\_declarations.csv — full FEMA declaration history
- us\_disasters\_m5.csv — filtered subset for modelling validation

Imported using Power BI Power Query.

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### Data Cleaning — Power Query Steps

Performed in Power Query Editor:

Steps for Data Cleaning

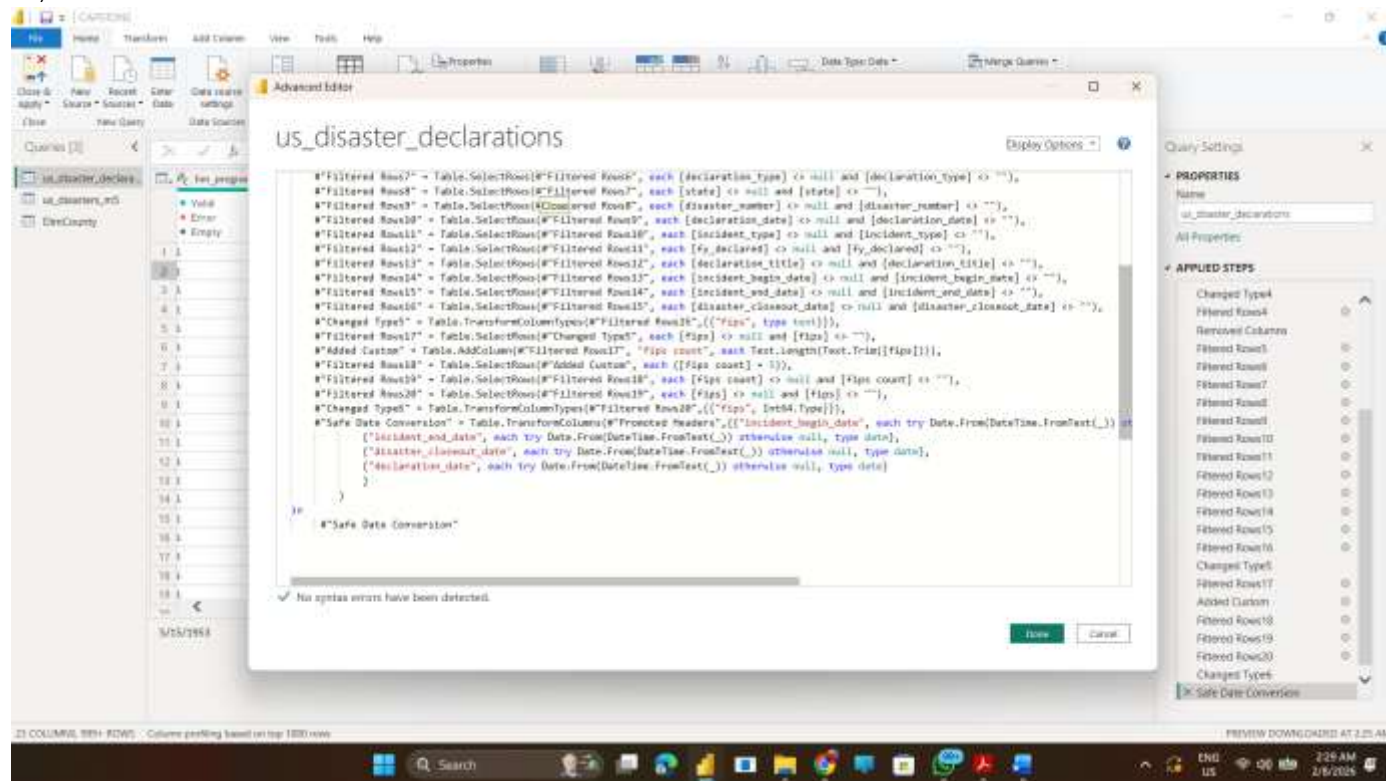
- Split date format column into date and time
- Check and remove blank values or missing values
- Change type of fips column to text
- Add custom column – use formula – “Text.Length(Text.Trim([fips]))”
- Filter the custom fips column to 5
- Change back the fips column to whole number
- Filtered null values and removed duplicates
- Handled malformed and missing FEMA date fields using defensive Power Query transformations to ensure reliable model refresh. – opened advanced editor in home tab then added this code in the end

```
#"Safe Date Conversion" =  
    Table.TransformColumns(  
        #"Promoted Headers",
```

```

{
  {"incident_begin_date", each try Date.From(DateTime.FromText(_)) otherwise null, type date},
  {"incident_end_date", each try Date.From(DateTime.FromText(_)) otherwise null, type date},
  {"disaster_closeout_date", each try Date.From(DateTime.FromText(_)) otherwise null, type date},
  {"declaration_date", each try Date.From(DateTime.FromText(_)) otherwise null, type date}
}
)

```



- Close and apply

### Fact Table

- Disaster\_Declarations (central fact)

### Dimension Tables

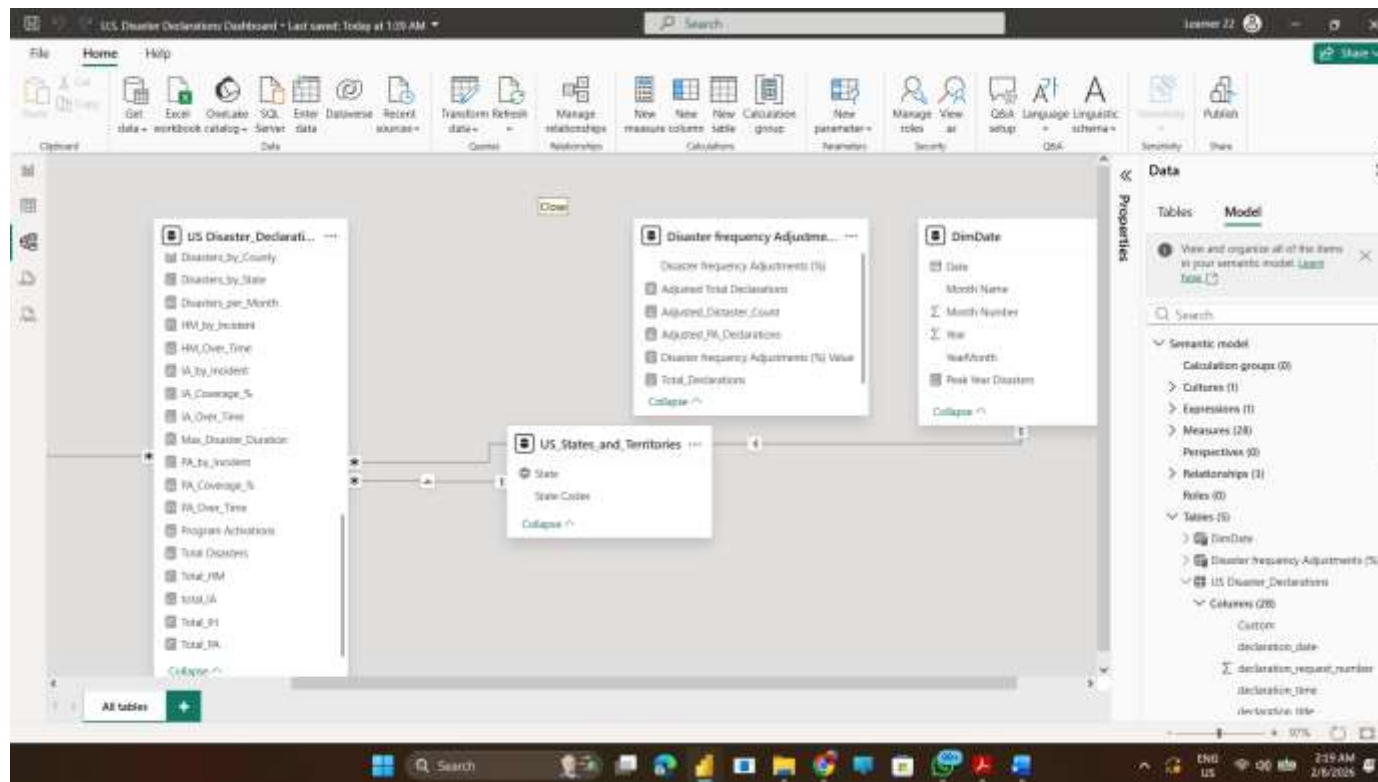
- DimDate
- US\_States\_and\_Territories
- Incident Type dimension
- Program indicators

### Relationships:

- DimDate → Fact (declaration\_date)
- State → Fact (state)
- Incident Type → Fact (incident\_type)

Model built using one-to-many relationships.

## ERD (Entity Relationship Diagram)



## DAX Measures Used in Dashboard

1. DimDate =

ADDCOLUMNS (

CALENDAR ( DATE(1953,1,1), DATE(2025,12,31) ),

```

"Year", YEAR([Date]),
"Month Number", MONTH([Date]),
"Month Name", FORMAT([Date], "MMM"),
"YearMonth", FORMAT([Date], "YYYY-MM")
)

```

2. Disaster frequency Adjustments (%) = GENERATESERIES(-0.2, 0.2, 0.2)

3. Adjusted Total Declarations = [Program Activations]\*(1+'Disaster frequency Adjustments (%)'[Disaster frequency Adjustments (%) Value])

4. Adjusted\_Disaster\_Count = [Total Disasters]\*(1+'Disaster frequency Adjustments (%)'[Disaster frequency Adjustments (%) Value])

**Purpose:** What-If scenario adjusted disaster count

**Used in:** Scenario Planning page

5. Adjusted\_PA\_Declarations = [Total\_PA]\*(1+'Disaster frequency Adjustments (%)'[Disaster frequency Adjustments (%) Value])

6. Disaster frequency Adjustments (%) Value = SELECTEDVALUE('Disaster frequency Adjustments (%)'[Disaster frequency Adjustments (%)], 0)

7. Total\_Declarations = [Total\_HM]+[total\_IA]+[Total\_PA]+[Total\_IH]

Or

Program Activations = [Total\_HM]+[total\_IA]+[Total\_PA]+[Total\_IH]

**Purpose:** Counts how many recovery programs were activated

**Used in:** Overview + Temporal Trends dual axis chart

8. Average\_Disaster\_Duration = AVERAGE('US Disaster\_Declarations'[Disaster\_Duration])

**Purpose:** Calculates average duration in days

**Used in:** Overview KPI + Incident Analysis

9. Average\_Time\_to\_Declaration = AVERAGE('US Disaster\_Declarations'[Time\_to\_Declaration])

10. Avg Disasters per Year = DIVIDE([Total Disasters], DISTINCTCOUNT(DimDate[Year]))

**Purpose:** Average number of disasters each year

**Used in:** Temporal Trends KPI card

11. Disaster\_Duration = DATEDIFF('US Disaster\_Declarations'[incident\_begin\_date], 'US Disaster\_Declarations'[incident\_end\_date], DAY)

12. Disasters\_by\_incident = DISTINCTCOUNT('US Disaster\_Declarations'[disaster\_number])

13. Disasters\_by\_County = COUNTROWS('US Disaster\_Declarations')

14. Disasters\_by\_State = DISTINCTCOUNT('US Disaster\_Declarations'[disaster\_number])

15. Disasters\_per\_Month = DISTINCTCOUNT('US Disaster\_Declarations'[disaster\_number])

16. HM\_by\_Incident = SUM('US Disaster\_Declarations'[hm\_program\_declared])

17. HM\_Over\_Time = SUM(('US Disaster\_Declarations'[hm\_program\_declared]))

18. IA\_by\_Incident = SUM('US Disaster\_Declarations'[ia\_program\_declared])

19. IA\_Coverage\_% = DIVIDE([total\_IA], [Total Disasters])

20. IA\_Over\_Time = SUM('US Disaster\_Declarations'[ia\_program\_declared])

21. Max\_Disaster\_Duration = MAX('US Disaster\_Declarations'[Disaster\_Duration])

22. PA\_by\_Incident = SUM('US Disaster\_Declarations'[pa\_program\_declared])

23. PA\_Coverage\_% = DIVIDE([Total\_PA], [Total Disasters])

24. PA\_Over\_Time = SUM('US Disaster\_Declarations'[pa\_program\_declared])

25. Time\_to\_Declaration = DATEDIFF('US Disaster\_Declarations'[incident\_begin\_date], 'US Disaster\_Declarations'[declaration\_date], DAY)

26. Total\_Disasters = DISTINCTCOUNT('US Disaster\_Declarations'[disaster\_number])

**Purpose:** Counts total disaster declaration records

**Used in:** Overview KPI cards, trend charts

27. Total\_HM = SUM('US Disaster\_Declarations'[hm\_program\_declared])

28. total\_IA = SUM('US Disaster\_Declarations'[ia\_program\_declared])

29. Total\_IH = SUM('US Disaster\_Declarations'[ih\_program\_declared])

30. Total\_PA = SUM('US Disaster\_Declarations'[pa\_program\_declared])

**Purpose:** Counts program declaration flags

**Used in:** Incident & Program Analysis page

31. Incident\_Group =

SWITCH(

TRUE(),

'US Disaster\_Declarations'[incident\_type] IN {

"Flood", "Flash Flood", "Coastal Storm",

"Dam/Levee Break", "Tsunami"

}, "Hydrological",

'US Disaster\_Declarations'[incident\_type] IN {

"Hurricane", "Tropical Storm", "Severe Storm(s)",

```
"Winter Storm", "Snowstorm", "Blizzard",  
"Ice Storm", "Hailstorm"  
, "Meteorological",
```

```
'US Disaster_Declarations'[incident_type] IN {  
"Earthquake", "Volcanic Eruption",  
"Landslide", "Mud/Landslide"  
, "Geological",
```

```
'US Disaster_Declarations'[incident_type] IN {  
"Fire", "Wildfire"  
, "Fire",
```

```
'US Disaster_Declarations'[incident_type] IN {  
"Drought", "Heat Wave", "Freezing"  
, "Climatological",
```

```
"Human / Other"
```

```
)
```

- The following DAX measures were created to support KPIs, trend analysis, scenario modeling, and geographic comparisons in the Power BI dashboard.



- Made a separate manual table “U.S.\_States and Territories” – State codes into State names

#### What-If Parameter Design

Created parameter:

Disaster Frequency Adjustment %

Range:-20% to +20%

Step: 5%

Used in DAX projected measures for scenario simulation.

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#### Optimization Steps

- Removed unused columns
- Used Top-N filters
- Reduced slicer cardinality
- Used star schema
- Hidden technical columns
- Disabled auto date/time

Refresh time < 3 minutes achieved.

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#### Dataset Update Procedure

When new FEMA data is published:

1. Replace CSV file with updated version

2. Open Power BI Desktop
3. Refresh Power Query
4. Validate date parsing
5. Refresh model
6. Republish to Power BI Service

#### **Scheduled Refresh Setup**

- Power BI Service:  
Workspace → Dataset → Settings → Scheduled refresh
  - Turn ON:  
Refresh = On  
Frequency = Monthly
  - Time = Choose slot
- Credentials → set data source credentials.